

The Grinder: A HANDY PERFORMANCE Testing Tool

In today's world of e-business, e-governance, e-revenue and e-commerce, performance testing helps identify bottlenecks, improves response times, sets fault tolerance and helps to recover from failures. This article will help Web testers to install and use The Grinder testing tool. It also promotes the use and development of open source performance testing tools.

A few days ago, I was working on some performance testing tools to find out which one to use to load test our software. My search ended with The Grinder, which uses Jython to make the test scripts that can be used to test applications.

The Grinder is an open source testing framework that allows the running of tests in parallel on many machines. It is a Java based framework and is available under a BSD style open source licence.

The key features of The Grinder are:

1. It uses a TCP proxy to record network activities.
2. It uses Jython (the power of Python combined with any Java API) to create test scripts.
3. It supports multiple protocols.
4. It provides support for distributed load testing.

The table below summarises these features.

Tool	The Grinder
OS Support	Any
GUI	Console based
Test recorder	TCP (including HTTP)
Test language	Python and Clojure
Extension language	Python and Clojure
Load report	Console
Host monitoring	No
Protocol	HTTP, SOAP, JDBC, POP3, SMTP, LDAP and JMS
Limitation	Knowledge of Python is required for test development. Reports are very simple and brief

In this article, readers will learn how to install and set up The Grinder, as well as record a test, modify it and start running the test.



Installation

I am using a Windows environment to install The Grinder version 3. You can run The Grinder on a Linux environment as well. The set-up is not much different for a Linux environment, the only differences being in the shell scripts and installation directory.

First check whether Java is installed on the machine. If not, please download and install Java from Oracle. Then follow the steps given below.

1. Download The Grinder from Sourceforge. The URL for the download is <http://sourceforge.net/projects/grinder/>.
2. Download the binary distribution, if you are planning to extend The Grinder, and then download the source distribution.
3. Unzip the binary file to some suitable location. I extracted the file to C:\Testing.
4. Download Jython2.1.jar class file from <http://www.jython.org/downloads.html>.

After downloading it, either run Java with the `-jar` option, as shown below:

```
java -jar jython_installer-2.2.1
```

Or double click `jython_installer-2.2.1.jar`

5. Add `grinder.jar` to the CLASSPATH:

```
Set CLASSPATH = %CLASSPATH%; C:\Testing\grinder-3.11-binary\grinder3.11\lib\grinder.jar
echo %CLASSPATH%
```

The Grinder has been successfully installed on your computer. You can check it by running The Grinder console. To do so, enter the following command in The Grinder directory: